

LeetCode 438: Find All Anagrams in a String

Topic: Fixed Window Frequency Map (Sliding Window)

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Problem Statement

Given two strings **s** and **p**, return an array of all the start indices of **p**'s anagrams in **s**. You may return the answer in any order.

Example:

```
s = "c b a e b a b a c d"
Indices: 0 1 2 3 4 5 6 7 8 9
```

```
p = "a b c"
```

Output: [0, 6]

Explanation: The substring with start index = 0 is "cba", which is an anagram of "abc". The substring with start index = 6 is "bac", which is an anagram of "abc".

Optimal Approach

- **Strategy:** Maintain a frequency array of size 26 for the characters in string **p**.
- For example, if **p = "abc"**, the frequency map will require: **a = 1, b = 1, c = 1**.

- **Sliding Window:** Slide a window of fixed size (equal to **p**'s size) over the string **s**.
- As you slide, update the frequency map (decrement for incoming characters, increment for outgoing characters).
- If at any point all values in the frequency array become 0, it means the current window is an exact anagram of **p**. Store the starting index of this window.

C++ Implementation

```

class Solution {
public:
    vector findAnagrams(string s, string p) {
        int n = s.size();
        int m = p.size();
        vector ans;
        if (m > n)
            return ans;

        vector freq(26, 0);

        for (char c : p) {
            freq[c - 'a']++;
        }

        for (int i = 0; i < m; i++) {
            freq[s[i] - 'a']--;
        }

        bool flag = true;
        for (int v = 0; v < 26; v++) {
            if (freq[v] != 0) {
                flag = false;
                break;
            }
        }

        if (flag)
            ans.push_back(0);

        for (int j = m; j < n; j++) {
            freq[s[j] - 'a']--; //aane waale frequency kam krenge
            freq[s[j - m] - 'a']++; //chala gaya to wapas frequency badha
do
            bool flag = true;
            for (int v = 0; v < 26; v++) {
                if (freq[v] != 0) { //kabhi bhi non zero dikhe matlab
equal freq nhi thi,cant add to ans
                    flag = false;
                    break;
                }
            }

            if (flag)
                ans.push_back(j - m + 1);
        }
    }
}

```

```
        return ans;  
    }  
};
```